

Examining Properties of Selective Divergence

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WHERE DISCOVERIES BEGIN

Core Notions

Definition (Guerrero Sánchez & Tkachuk 2017)

A space X is said to be *discretely selective* if, for any sequence $\langle U_n : n \in \omega \rangle$ of nonempty open subsets of X , there exists $\langle x_n : n \in \omega \rangle$ such that $x_n \in U_n$ for each $n \in \omega$ and $\{x_n : n \in \omega\}$ is closed and relatively discrete in X .

Definition (Jiménez-Flores, Ríos-Herrejón, Rojas-Sánchez, & Tovar-Acosta 2026)

A space X is said to be *selectively highly divergent* (SHD) if, for any sequence $\langle U_n : n \in \omega \rangle$ of nonempty open subsets of X , there exists $\langle x_n : n \in \omega \rangle$ such that $x_n \in U_n$ for each $n \in \omega$ and the sequence $\langle x_n : n \in \omega \rangle$ has no convergent subsequence.

Today's Main Message

The properties of being discretely selective and of being SHD are generally distinct in the class of all topological spaces and groups, but agree in spaces $C_p(X)$ and $C_k(X)$.

- In function spaces, the selection process can be guaranteed to form an injection, inspiring the property of being *injectively* discretely selective.

Continuous Function Spaces

- $C_p(X)$, the space of continuous functions $X \rightarrow \mathbb{R}$ with the topology of point-wise convergence; basic neighborhoods:

$$[f; F, \varepsilon] = \{g \in C(X) : (\forall x \in F) |f(x) - g(x)| < \varepsilon\}$$

- $C_k(X)$, the space of continuous functions $X \rightarrow \mathbb{R}$ with the topology of uniform convergence on compact subsets; basic neighborhoods:

$$[f; K, \varepsilon] = \{g \in C(X) : (\forall x \in K) |f(x) - g(x)| < \varepsilon\}$$

Relevant Hyperspaces

We will also explore discrete selectivity and SHD in hyperspaces of finite subsets.

- $\mathcal{F}[X]$, the Pixley-Roy hyperspace of nonempty finite subsets of X ; basic neighborhoods:

$$[F, U] = \{G \in [X]^{<\aleph_0} : F \subseteq G \subseteq U\}$$

- $\mathcal{P}_{\text{fin}}(X)$, the Vietoris hyperspace of nonempty finite subsets of X ; basic neighborhoods:

$$\begin{aligned} & [U_1, U_2, \dots, U_n] \\ &= \left\{ F \in [X]^{<\aleph_0} : F \subseteq \bigcup_{j=1}^n U_j \wedge \bigwedge_{j=1}^n F \cap U_j \neq \emptyset \right\} \end{aligned}$$

Basic Observations

- Any discrete space is discretely selective but not SHD (also not injectively discretely selective).
- No infinite Hausdorff countably compact space is discretely selective (but some are SHD).

Fact (JRRTT 2024)

If X is not sequentially compact and κ is uncountable, X^κ is SHD.

- $(\beta\omega)^{\omega_1}$ is SHD but not discretely selective.
- $\mathbb{T}^c$, where $\mathbb{T}$ is the circle group, is a connected group which is SHD but not discretely selective.
- $2^\mathfrak{s}$, where $\mathfrak{s}$ is the splitting number, is a zero-dimensional group which is SHD but not discretely selective. (Bella, Spadaro 2024)

Question

Is there a crowded Hausdorff space (group) which is discretely selective but not SHD?

Theorem (Bella, Spadaro 2024)

If X is SHD, then so is $\mathcal{F}[X]$.

- $\mathcal{F}[\Sigma(2, 0, \omega_1)]$ is SHD, but $\Sigma(2, 0, \omega_1)$ is not. (JRRTT 2024)
 - $\Sigma(2, 0, \omega_1) = \{f \in 2^{\omega_1} : |f^{-1}(1)| \leq \aleph_0\}$
 - The proof provided also shows that $\mathcal{F}[\Sigma(2, 0, \omega_1)]$ is injectively discretely selective.
 - Since $\Sigma(2, 0, \omega_1)$ is an infinite Hausdorff sequentially compact space, $\Sigma(2, 0, \omega_1)$ is not discretely selective.

Theorem (C. 2026)

If X is SHD, then so is $\mathcal{P}_{\text{fin}}(X)$.

Question

If $\mathcal{P}_{\text{fin}}(X)$ is SHD, is X necessarily SHD?

Theorem (C. 2026)

If X is discretely selective, then so is $\mathcal{P}_{\text{fin}}(X)$.

In fact, we have a full game equivalence between the closed discrete selection game on X and a game on $\mathcal{P}_{\text{fin}}(X)$ where the winning condition for the second player in $\mathcal{P}_{\text{fin}}(X)$ is that $\{F_n : n \in \omega\}$ is not only closed and discrete in $\mathcal{P}_{\text{fin}}(X)$, but also that $\bigcup\{F_n : n \in \omega\}$ is a closed discrete subset of X .

Question

If $\mathcal{P}_{\text{fin}}(X)$ is discretely selective, is X necessarily discretely selective?

When is $\mathcal{F}[X]$ discretely selective?

- If X is discrete, then so is $\mathcal{F}[X]$.
- Otherwise, determining whether $\mathcal{F}[X]$ is discretely selective seems to be a bit more complicated.

Definition

A space is *nowhere separable* if no nonempty open subset is separable, or $mld(X) > \aleph_0$.

Density: $d(X) = \min \{ |A| : A \subseteq X, \overline{A} = X \}$

$$mld(X) = \min \left\{ d(U) : \emptyset \neq U \underset{\text{open}}{\subseteq} X \right\}$$

minimal local density?

Theorem (C. 2026)

If X is T_1 , injectively discretely selective, and nowhere separable, then $\mathcal{F}[X]$ is (injectively) discretely selective.

- $\mathbb{Z}^\kappa$, $\kappa \geq \mathfrak{c}^+$, is a nowhere separable T_1 topological group which is injectively discretely selective.
- 2^κ , $\kappa \geq \mathfrak{c}^+$, is a nowhere separable T_1 topological group which is not discretely selective.

Question

If X is discretely selective, is $\mathcal{F}[X]$ necessarily discretely selective?

A Weakening

Definition (Jiménez-Flores, Ríos-Herrejón, Rojas-Sánchez, Tomita, & Tovar-Acosta 2024)

A space X is said to be *weakly selectively highly divergent* (WSHD) if, for any sequence $\langle U_n : n \in \omega \rangle$ of nonempty open subsets of X , there exists $\langle x_n : n \in \omega \rangle$ such that $x_n \in U_n$ for each $n \in \omega$ and the sequence $\langle x_n : n \in \omega \rangle$ has **a subsequence for which no further subsequence is convergent**.

A similar modification can be made for discrete selectivity, as well as other selective properties and games.

Question

Is WSHD equivalent to SHD?

What about the analogous question for discrete selectivity?

Theorem (JRRTT 2024)

If X is a regular P -space and $\mathcal{F}[X]$ is WSHD, then $\mathcal{F}[X]$ is SHD.

Theorem (JRRTT 2024)

If X is a Tychonoff space and $\mathcal{F}[C_p(X)]$ is WSHD, then $\mathcal{F}[C_p(X)]$ is SHD.

Theorem (JRRTT 2024 / C. 2026)

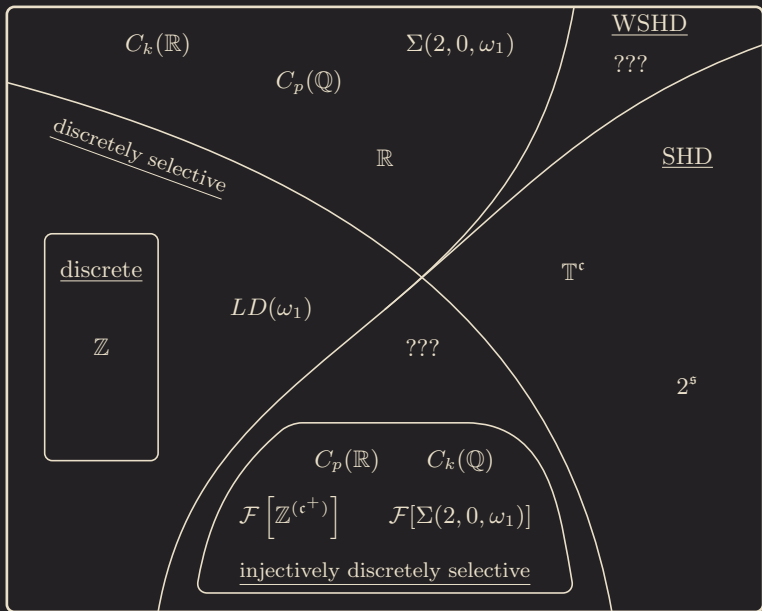
For a Tychonoff space X , the following are equivalent:

- X is uncountable;
- $C_p(X)$ is injectively discretely selective;
- $C_p(X)$ is discretely selective;
- $C_p(X)$ is SHD;
- $C_p(X)$ is WSHD.

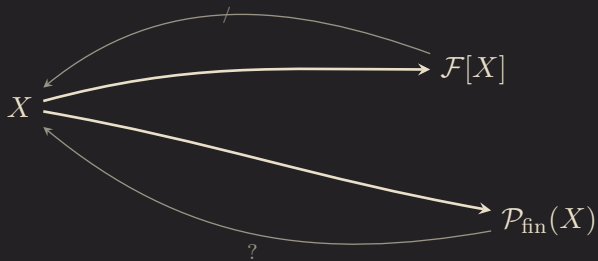
Theorem (C. 2026)

For a Tychonoff space X , the following are equivalent:

- X is not hemicompact;
- $C_k(X)$ is injectively discretely selective;
- $C_k(X)$ is discretely selective;
- $C_k(X)$ is SHD;
- $C_k(X)$ is WSHD.



SHD



discretely
selective

